Aswini Kumar Padhi

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Summary

I am an applied NLP researcher with 7+ years of experience specialized in translating complex unstructured data into actionable NLP pipelines and scalable machine learning solutions with domain-centric language modeling techniques. I have collaborated and led research partnerships across industry and academia, delivering 6 high-impact projects along with top-tier research publications.

Education

Indian Institute of Technology, Delhi — New Delhi, India
Ph.D. in Computer Science (CGPA: 9.0/10*)* — *June 2023 – Present*

Indian Institute of Information Technology — Gwalior, India
Mtech in Computer Science (CGPA: 9.38/10*) (Department Gold Medalist)* — *Aug. 2021 – May 2023*

Veer Surender Sai University of Technology — Burla, India
Btech in Computer Science (CGPA: 8.4/10*)* — *Aug. 2012 – May 2016*

Work Experience

Doctoral Researcher — June 2023 – Present
Indian Institute of Technology, Delhi (LCS2 lab) — *New Delhi, India*

- **Focus Area:** Agentic LLMs for Evaluating and Mitigating Online Hate.
- Curated 6 high quality hate-counter datasets spanned over 10 different target groups across countries for evaluating the impact of online hate and modeling effective mitigation strategies.
- Developed 5 different novel parameter efficient reinforcement learning frameworks and 2 different novel optimized fine-tuning to align LLM agents towards human values.

Visiting Researcher — Feb. 2026 – May 2026
Indian Institute of Technology, Delhi, Abu Dhabi — *Abu Dhabi, UAE*

- Led an internal collaborative research utilizing mechanistic interpretability to localize and mitigate bias, resolving underlying model failures to achieve an 18% improvement in unbiased response generation across 5 foundational LLMs.

Visiting Researcher — Dec. 2025 – Jan 2026
University of Minnesota — *Minneapolis, US*

- Led DST-NSF collaborative research assessing digital vulnerabilities to misinformation and malinformation, deploying robust LLM-driven mitigation strategies to enhance online safety.

Research Internship — Sep. 2025 – Dec. 2025
Microsoft Research India — *Bengaluru, India*

- Developed novel agentic evaluation strategies and evaluated 5 distinct LLM agentic frameworks across 15 novel evaluation verticals, developing a diagnostic pipeline that systematically quantified reasoning breakdowns and isolated critical bottlenecks during real-world execution.

Research Engineer — June 2023 – May 2025
Logically AI (Onsite) — *New Delhi, India*

- **Benchmarking and Evaluation:** Engineered a high-fidelity CS-eval dataset spanning 4 distinct verticals, establishing a robust benchmark for toxicity mitigation that strongly correlates with human alignment judgments.
- **Mitigation Frameworks:** Curated 2 large-scale, high-quality datasets and pioneered 2 novel automated mitigation methodologies to systematically detect and neutralize harmful online content.

Associate Consultant on Machine Learning for VALE S.A. — July 2019 – Dec 2020
Wipro Technologies — *Hyderabad, India*

- **Job Role:** Inventory Optimization for Iron Ore Supply Chain through Machine Learning
- worked on designing TensorFlow-based deep learning models in collaboration with global logistics teams, leveraging historical shipment data to forecast iron ore demand across international markets.

- Deployed the forecasting pipeline directly into VALE's ERP system via API integrations, incorporating macroeconomic indicators and market pricing to drive real-time inventory optimization.

Associate Consultant on Machine Learning for Chevron USA Dec. 2016 – June 2019
Wipro Technologies *Bengaluru, India*

- **Job Role:** Machine Learning-Based Predictive Maintenance System for Oil Equipment
- Directed the remote development of a predictive maintenance pipeline for Chevron, engineering Random Forest models (scikit-learn) to analyze IoT sensor telemetry from 1,000+ rotating assets.
- Collaborated with on-site engineering teams to extract predictive features from vibration, temperature, and pressure fluctuations, establishing an early-warning system for mechanical failures.

Technical Skills

- **Data Pipeline:** Text scraping, quality assurance, annotation design, and evaluation
- **ML/NLP:** Multi-Agent modeling, fine-tuning, parameter-efficient fine-tuning, reinforcement learning, prompt engineering, diffusion language modeling
- **Evaluation:** Significance testing, error analysis, LLM-as-a-Judge
- **Tools:** Python, PyTorch, Hugging Face, Pandas, Git

Soft Skills

- **Collaboration:** Scientific communication, international and cross-disciplinary teamwork, critical thinking

Major Publications

See Google Scholar for complete list ** Equal contribution*

Parameter-Efficient LLM Learning Dynamics

- **[ACL 2025]** *Counterspeech the ultimate shield! Multi-Conditioned Counterspeech Generation through Attributed Prefix Learning* **A. Padhi**, A. Bandhakavi, and T. Chakraborty.
- **[NAACL 2024]** *Intent-conditioned and Non-toxic Counterspeech Generation using Multi-Task Instruction Tuning with RLAIF.* Amey Hengle*, **A. Padhi***, S. Singh, A. Bandhakavi, S. Akhtar, and T. Chakraborty.

Domain-Specific Evaluation

- **[NAACL 2024]** *CSEval: Towards Automated, Multi-Dimensional, and Reference-Free Counterspeech Evaluation using Auto-Calibrated LLMs.* A. Hengle*, **A. Padhi***, A. Bandhakavi, and T. Chakraborty.

Miscellaneous

- **Honors & Fellowship:** Winner Microsoft Research India PhD Fellow (2025), Finalist Qualcomm Innovation Fellowship 2025 India, Department gold medal received from Honorable President of India (2023).
- **Invited Talks:** Microsoft Research India Academic Summit (2025, 2026), ACM ARCS India (2026), IIIT Gwalior, India (2025), IIIT Jabalpur, India (2024).
- **Reviewer Services:** Reviewer for ML/NLP venues - ACL, EMNLP, EACL, AACL, NAACL.
- **Mentorship:** Supervised 3 master students, 2 bachelor students and 2 research assistants.